SIP Products

TP-260/SIP
PCI form-factor digital gateway, available in 1, 2, 4, and 8 digital T1/E1 spans

IPM-260/SIP
PCI form-factor media Server platform, available in 30-240 Media Resources, 8 E1/T1 spans

Mediant™ 2000
1 to 16 span digital media gateway for enterprise & carrier applications

IPmedia™ 2000
Media server platform providing conferencing, transcoding and tone detection resources

MediaPack™ Series
Analog media gateways with 4-24 port (FXS) or 4-8 port (FXO) connectivity

AC494 Voice over Packet SoC
System on a Chip family for IP phone and CPE developers

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Your Gateway to VoIP

Everything You Wanted to Know About

SIP

AudioCodes

AudioCodes SIP Partners

abp Technology Partners
AVAYA
BROADSOFT
digium | Asterisk
eDial
FaxBack
GENESYS AN ALCATEL COMPANY
IBM Business Partner
INTERACTIVE INTELLIGENCE
natural convergence
pingtel
Sonus NETWORKS
Sphere
Swork
VocalData
VONEXUS

SIP Terminology

SIP: Session Initiation Protocol (RFC 3261) – Application-layer control (signaling) protocol for creating, modifying, and terminating sessions with one or more participants.

SIP Methods: SIP protocol commands or messages (eg: INVITE, BYE)

SIP Response Codes: Responses to SIP methods indicating success, failure or other information. (eg: 200 – Ok)

SIP User Agent (UA): An endpoint device that can issue or respond to SIP protocol methods.

SIP User Agent Client (UAC): A SIP endpoint device issuing the request (eg: Phone, PC, PDA...).

SIP Server (UAS): An application or embedded software that can accept and respond to SIP methods.

SIP Gateway: A network element that can convert SIP methods and response codes to another protocol.

SIP Proxy Server: An intermediary entity that acts as both a server and a client for the purpose of making requests on behalf of other clients.

SDP: Session Description Protocol (RFC 2327) – Text-based protocol describing multi-media sessions.

Softswitch: Software application that coordinates VoIP call switching between endpoints, commonly duplicating

Helpful URLs

www.ietf.org RFC 3261 – The current official specification
www.cs.columbia.edu/sip/ Excellent and well organized reference materials
www.packetcomm.org Packet Communications Forum

SIP Methods

REGISTER: Registers a user with a Proxy/Registrar

INVITE: Session setup request or media negotiation. Used also to hold & retrieve calls

CANCEL: Used to cancel an INVITE transaction

ACK: Acknowledgement for an INVITE transaction completion

BYE: Terminating a session

OPTIONS: Used as a query for remote's status & capabilities

INFO: Mid-call signaling information exchange

SUBSCRIBE: Request notification of call events

NOTIFY: Event notification after an explicit/implicit subscription

REFER: Call Transfer request

SIP Response Codes

100: Trying – Request has been received by a proxy/gateway

180: Ringing – The called party received the INVITE request, the phone is ringing.

181: Call is being forwarded

182: Queued – Invite has been received and will be processed in a queue

183: Session Progress – Used to convey report of incoming early-media

200: OK - successful transaction completion

302: Moved Temporarily – Forward call to a given contact

305: Use Proxy – Repeat same call setup using a given proxy

400: Bad Request – General error

401: Unauthorized – The server requires client authentication

404: Not Found – The user does not exist at the specified domain

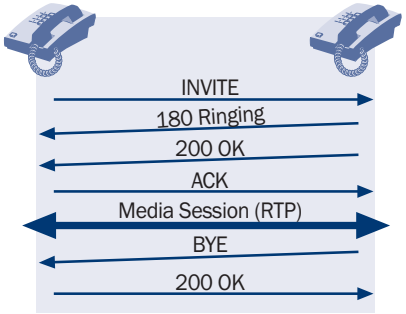
408: Request Timeout

486: Busy here

5xx: Server Failure

6xx: Global Failure

Basic SIP Call Example



INVITE Message in Detail

INVITE sip:Bob@Broadway.com SIP/2.0
Via: SIP/2.0/UDP
audiocodes.com:5060;branch=z9hG4bK74bf9
From: Alice <sip:Alice@audiocodes.com>;tag=1c289323
To: Bob <sip:Bob@audiocodes.com>
Call-ID: abfd43978343@audiocodes.com
Max-Forwards: 70
CSeq: 1 INVITE
Contact: Alice <sip:Alice@audiocodes.com>
Content-Type: application/sdp
Content-Length: 142

SDP Content.
Lists Alice's supported coders, IP Address & Port for voice media

v=0
o=Alice 7439443843 7439443843 IN IP4 audiocodes.com
s=-
c=IN IP4 198.64.138.249
t=0 0
m=audio 10000 RTP/AVP 0
a=rtpmap: 0 PCMU/8000

SIP Fields

Field	Meaning
INVITE header	Inviting user at SIP address Bob@Broadway.com to a media session.
Via	The response to the INVITE message should be returned to the specified address, using the specified protocol (UDP). The SIP protocol is carried over UDP, TCP, STCP & TLS. The default SIP ports are 5060 for UDP, TCP, SCTP and 5061 for TLS.
From	The calling party. The caller can choose to remain anonymous by filling the address field with "anonymous". A unique tag is created by the initiating party to help identify the addressee in future messages.
To	The called party. In later responses, the called party should add its own unique tag, similar to the one represented with the From party.
Call-ID	A unique field, used to identify the call.
Max-Forwards	The Max-Forwards value is an integer in the range 0-255 indicating the remaining number of times this request message is allowed to be forwarded.
CSeq	Command Sequence header. The field value advances with each new message. Responses carry the same CSeq as their corresponding Requests.
Contact	The SIP Address of the calling party. In short, how the called party should reach the calling party in future messages.
Content-Type	SIP messages carry bodies that are transparent to the SIP Protocol. The Content-Type distinguishes one body type from another. In this example SDP is being carried. The media session is being negotiated using the SDP.
Content-Length	The length of the body (in bytes). Explicitly announced for technical reasons.
SDP	Session Description Protocol. Used to announce the coder capabilities, media IP address & ports.